

BRAIN METABOLISM

Brain is the most metabolically active organ in the body, despite it represents only 2% of body weight. It consumes about 20% of the body resting energy and 20% of the body oxygen supply

➤ **Fuels used by brain:**

1- Glucose. 2- Ketone bodies. 3- Amino acid 4- Lactic acid

1-Glucose:

- The brain relies almost exclusively on glucose for energy under normal condition
- Nervous system closely depends on the availability of **glucose**, the lack of which quickly results in death.
- **Glucose** is brought to the brain by blood flowing through a special network of arteries and is quickly catabolized by the **glycolytic** and **tricarboxylic acid cycle (Kreb's cycle)** pathways to synthesize **energy**.
- It is also used in the synthesis of amino acids, nucleotides and NADPH (HMP pathway).
- **Glucose** can be converted into **glycogen** in the brain.
- The astrocyte is capable of synthesizing glucose-6-phosphate in the **gluconeogenic pathway**
- Brain carbohydrate metabolism is thus in many ways similar to liver carbohydrate metabolism.
- **The astrocytes** are the principal brain cells responsible for many aspects of brain carbohydrate metabolism, rather than neurons.

➤ **Brain Energy Metabolism:**

- The **brain** stores little amount as glycogen and relies almost entirely on circulating glucose for fuel.
- Inside neurons, glucose is metabolized by cytoplasm & mitochondria in a number of steps (**glycolysis & Krebs's cycle**) to produce cellular **energy**, as **ATP**.

➤ **Role of Oxygen to Brain:**

- Having enough **oxygen** in the **brain** is crucial to brain function, and therefore, to the body as a whole.
- The **brain** requires a constant supply of both **oxygen** and **glucose**.
- Some brain cells start dying less than 5 minutes after their oxygen supply disappears.
- Without enough **oxygen** or with **low blood-oxygen** levels, lead to **brain** death.

➤ **Importance of Glucose to the brain:**

1. Major source of energy for RBC and CNS.
2. Neurotransmitter Synthesis as GABA and Acetylcholine
3. Maintenance of Brain Function (Adequate glucose levels are essential for normal cognition, memory, and consciousness)
4. Hypoglycemia can cause confusion, dizziness, seizures and coma

2-Ketone Bodies:

- **Metabolic Switching:** In prolonged fasting (greater than 2–3 weeks), periods of intense carbohydrate restriction, or medically induced starvation; **ketone bodies** (like beta-hydroxybutyrate) can partially replace glucose
- Ketones can provide up to 60% of the brain's energy needs during extended periods without glucose.

3-Amino Acids :

- The three **amino acids** with the highest **levels in brain tissues** are **glutamic acid, glutamine, and aspartic acid.**
- During starvation muscle proteins are broken down, releasing amino acids.
- These amino acids are transported mainly to the liver, where they are used through gluconeogenesis to produce glucose.
- The brain then uses the newly synthesized glucose rather than the amino acids themselves.

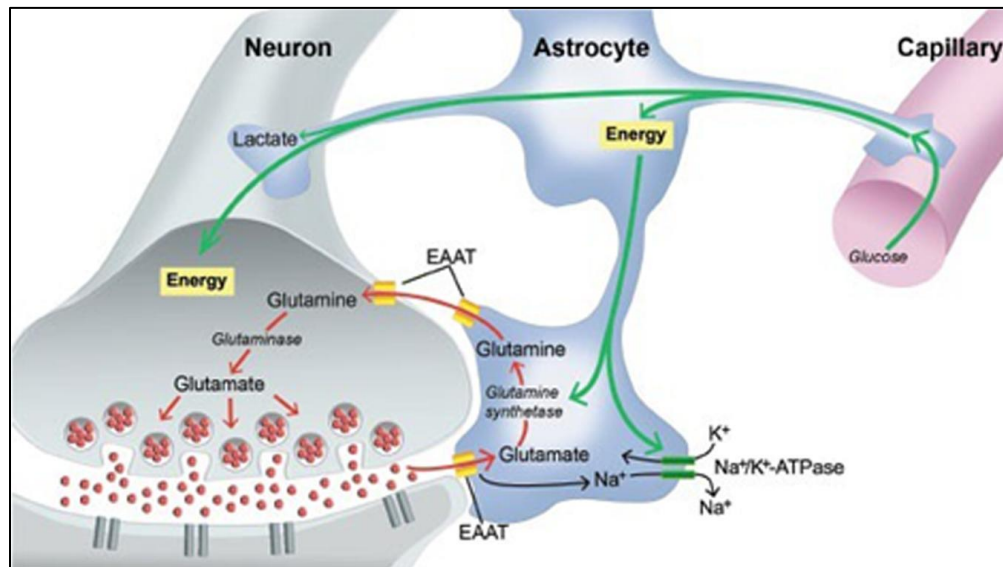
Glucogenic Amino Acids

Glucogenic amino acids are amino acids that can be converted into converted into glucose through gluconeogenesis. (ex: alanine, aspartic, glutamic, glycine...)

The Glutamate-Glutamine Cycle:

- **Glutamate** is the main excitatory neurotransmitter in the brain but too much glutamate outside neurons can become toxic and damage nerve cells.
- Astrocytes remove excess glutamate from around neurons and convert it into glutamine.

- Glutamine is transferred back to neurons, where it is converted again into glutamate for neurotransmission or used in cellular metabolism.
- During amino acid metabolism, toxic ammonia is produced.
- Astrocytes detoxify ammonia by incorporating it into glutamine, helping protect brain tissue from ammonia toxicity.
- **SO:** astrocytes protect the brain by recycling glutamate and detoxifying ammonia through the formation of glutamine.



4- Lactic Acid :

The brain can use lactic acid by taking it up directly from the bloodstream or receiving it from supporting cells astrocyte

a- Blood-Borne (Systemic) :

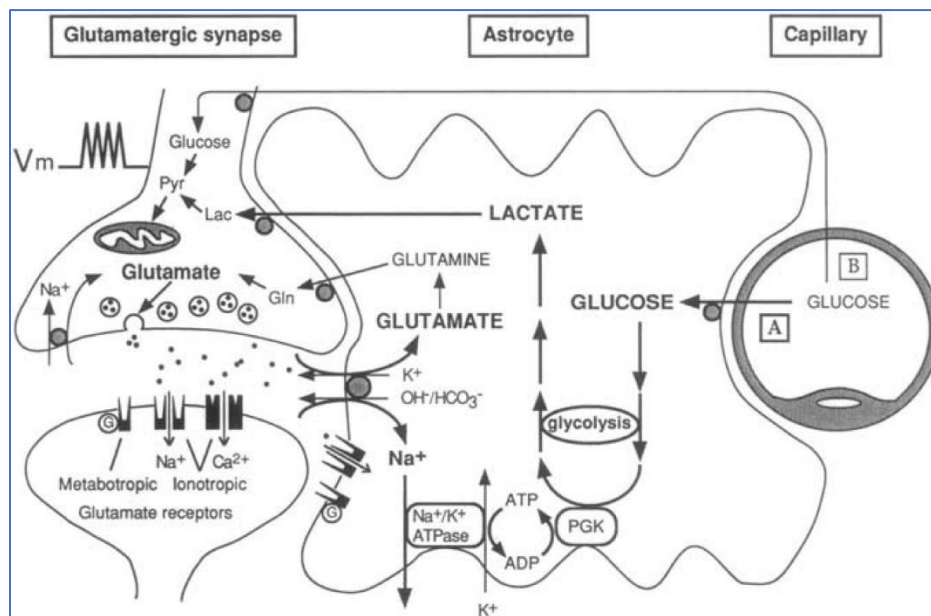
During high-intensity physical exercise or periods of low blood sugar, the muscles and other tissues release large amounts of lactate into the blood. The brain's uptake of lactate increases significantly, and it readily oxidizes this circulating lactate for energy

b-The Astrocyte-Neuron Lactate Shuttle (ANLS) Production:

When the brain is active, astrocytes (glial cells) take up glucose from the blood and break it down via glycolysis, producing lactate.

Delivery: Astrocytes then release this lactate, which is transported into active neurons to meet their sudden, high energy demands.

Consumption: Neurons convert this lactate into pyruvate by enzyme lactate dehydrogenase (LDH) then burn it aerobically to produce rapid energy



➤ **Fed Starvation Cycle**

Fate of glucose in Fed State

Fate of glucose in liver:

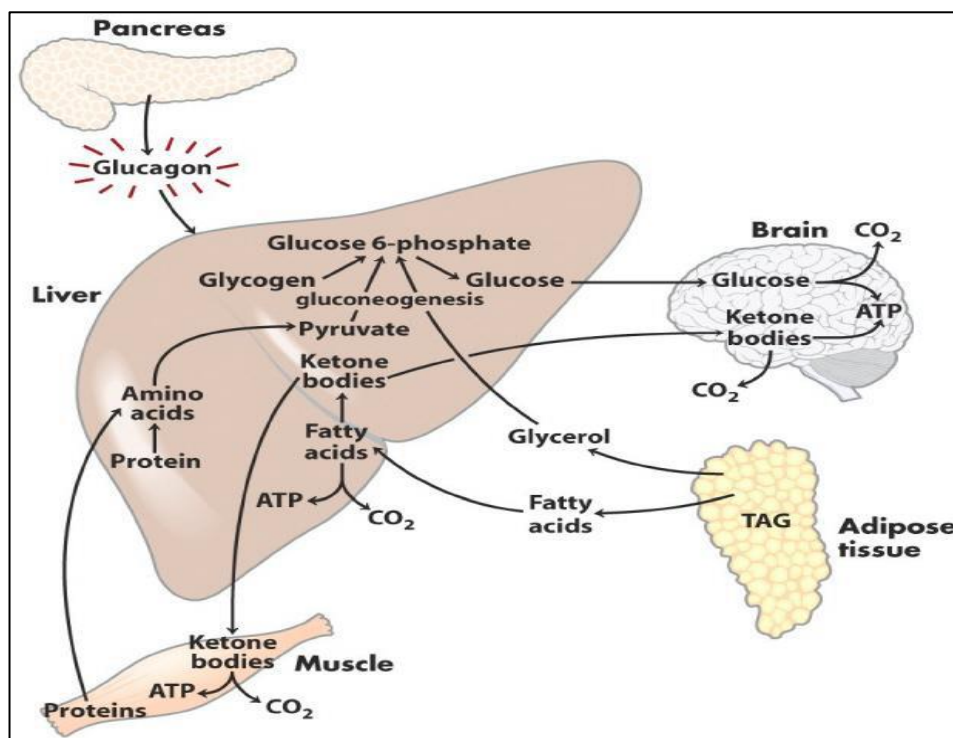
A- Used for **Glycolysis** then **Kreb's Cycle**

B-**Excess glucose:** stored in liver as **Glycogen** which used to maintain blood glucose between meal

C-More **excess glucose:** converted to **TriAcylGlycerol (TAG)** synthesis

Fate of glucose in other tissues

- **Brain:** make **Glycolysis** then **Kreb's Cycle**
- **RBC:** make **anaerobic Glycolysis** end by lactic acid
- **Muscle:**
 - 1- used for **Glycolysis** then **Kreb's Cycle**
 - 2- Excess glucose: stored in muscle as **Glycogen**



➤ Brain in Starvation

- During first few day of starvation brain use available glucose exclusively and can use glucose that provided from rapid breakdown of **muscle protein**
- After 3 week of starvation **ketone bodies** markedly increase and used as fuel for brain, this decreases breakdown of muscle protein

Clinical applications

1. Ammonia intoxication

It is the accumulation of ammonia (NH_3) in the blood and brain to toxic levels. It commonly

Symptoms:

- a) Flapping tremor
- b) Slurring of speech
- c) Blurring of vision
- d) Vomiting in infant
- e) Coma and death

Types

1- Acquired hyperammonemia

Due to liver cirrhosis

2- Inherited hyperammonemia

Due to urea cycle enzyme deficiencies

Why ammonia is toxic to the brain

- a) **Excess glutamine formation in astrocytes** causes astrocyte swelling and cerebral edema.
- b) **Depletion of α -ketoglutarate** reduces TCA cycle activity and decreased ATP production
- c) **Decreased glutamate levels** decreases GABA metabolism (an inhibitory neurotransmission).
- d) **Increased oxidative stress** damages neurons.
- e) **Disturbed ion balance and neurotransmission** impair brain function.

2. Cerebral insufficiency:

Also known as cerebrovascular insufficiency or cerebral hypoperfusion, refers to an inadequate supply of blood, oxygen, or nutrients to the brain.

This deficiency can be acute (like a stroke) or chronic.

a. Acute Cerebral Insufficiency:

Characterized by sudden drops in blood flow resulting in an ischemic stroke or a Transient Ischemic Attack (TIA).

b. Chronic Cerebral Circulation Insufficiency (CCCI):

A long-term reduction in blood flow, usually caused by arteriosclerosis (narrowing of the arteries). It is a major risk factor for vascular dementia.

3. Vertebrobasilar Insufficiency (VBI):

A specific type of insufficiency that affects the posterior circulation, which controls vision, balance, and coordination. It is primarily driven by progressive vascular and cardiovascular diseases rather than sudden acute events.

Causes and Risk Factors

The condition is usually a result of blocked, narrowed, or diseased blood vessels leading to or inside the brain.

The most prevalent causes are:

- a) Atherosclerosis:** Buildup of fatty plaques in the carotid or vertebral arteries.
- b) Medical:** High blood pressure, high cholesterol, diabetes
- c) Life style:** smoking, obesity.

Common Symptoms:

- a) Cognitive:** Memory loss, confusion, and difficulty concentrating.
- b) Physical:** Dizziness, vertigo, lightheadedness, and difficulty walking.
- c) Sensory/Motor:** Sudden muscle weakness (e.g., hemiplegia), numbness, or temporary loss of vision.